

Number System Cheat Sheet

1. Types of Numbers

Natural: 1,2,3... | Whole: 0,1,2... | Integers: -3 to +3 | Rational: p/q | Irrational: $\sqrt{2}$, π | Real: all

2. Divisibility Rules

Number	Rule
2	Last digit even
3	Sum of digits divisible by 3
4	Last 2 digits divisible by 4
5	Ends with 0 or 5
6	Divisible by 2 & 3
8	Last 3 digits divisible by 8
9	Sum divisible by 9
10	Ends with 0

3. Important Formulas

Sum of n numbers: $n(n+1)/2$

Sum of squares: $n(n+1)(2n+1)/6$

Sum of cubes: $[n(n+1)/2]^2$

$LCM \times HCF = a \times b$

4. Concepts

Even/Odd rules, Prime numbers (2 is only even prime), Co-prime ($HCF=1$)

5. Shortcuts

Use cyclicity for last digit, digital root for fast calculation

6. Important Problems

Remainder, Unit digit, LCM/HCF, Factors, Trailing zeros, Base conversion